

Name: _____

Date: _____

Solve each problem. Show your work in the box.

Write your final answer on the answer line (or in the blanks shown).

Use the strategies you learned in Lesson 5.11.07.

1 Rhombus Angles

A rhombus has one angle of 72° .

Find all four angles of the rhombus.

Opposite angles: 72° and 72° . Adjacent supplementary angles: $180^\circ - 72^\circ = 108^\circ$ each. Angles: 72° , 108° , 72° , 108° . Sum =

2 Trapezoid Co-interior Angles

In a trapezoid, the two angles on the same leg (co-interior angles) are supplementary. One base angle is 58° .

Find the co-interior angle on the same leg.

$180^\circ - 58^\circ = 122^\circ$. _____

3 Kite Angles

A kite has angles 80° , 80° , 120° , and an unknown angle.

Find the unknown angle.

$360^\circ - 80^\circ - 80^\circ - 120^\circ = 80^\circ$. _____

4 Multi-Step: Quadrilateral in a Diagram

A quadrilateral is divided into two triangles by a diagonal. Each triangle has an angle sum of 180° .

Use this fact to explain why the sum of angles in a quadrilateral is 360° .

Each triangle contributes 180° . Two triangles together = $2 \times 180^\circ = 360^\circ$, which equals the angle sum of the quadrilateral.

5 Reasoning: Mira's Pentagon Claim

Mira says that if a quadrilateral's angles sum to 360° , then a pentagon's angles must sum to 540° .

Is Mira correct? Explain by dividing a pentagon into triangles.

Yes. A pentagon can be split into 3 triangles. $3 \times 180^\circ = 540^\circ$. Mira is correct.